

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 07.2

Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;
% Signal
fs = 1000; N = 256; t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);
% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular', 'Hamming', 'Hanning'};
% Processing
for i = 1:3
    xw = x(:).*w{i};
    X = fft(xw,NFFT);
    P{i} = abs(X(1:NFFT/2));
    P{i} = P{i}/max(P{i});
end
figure;
for i = 1:3
    subplot(3,1,i);
    plot(f,20*log10(P{i}), 'LineWidth',1.5);
    title(['Spectrum using ' names{i} ' Window']);
    xlim([0 250]); grid on;
end
figure;
plot(f,20*log10(P{1}), 'LineWidth',1.5); hold on;
plot(f,20*log10(P{2}), 'LineWidth',1.5);
plot(f,20*log10(P{3}), 'LineWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on;
figure;
for i = 1:3
    subplot(3,1,i); plot(w{i}, 'LineWidth',1.5);
    title([names{i} ' Window']); grid on;
end
for i = 1:3
    [pks{i}, loc{i}] = findpeaks(P{i},f, 'MinPeakHeight',0.3);
    fprintf('\n%s Window Frequencies:\n',names{i});
    disp(loc{i});
end
figure;
plot(f,20*log10(P{2}), 'LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}), 'ro', 'MarkerFaceColor','r');
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;
fprintf('\nMain Spectral Peaks:\n');
for i = 1:3
    [~,idx] = max(P{i});
    fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));
end
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
end
```

```
%% Individual Spectrums
```

```
figure;
```

```
for i = 1:3
```

```
    subplot(3,1,i);
```

```
    plot(f,20*log10(P{i}), 'LineWidth',1.5);
```

```
    title(['Spectrum using ' names{i} ' Window']);
```

```
    xlim([0 250]); grid on;
```

```
end
```

```
%% Comparison
```

```
figure;
```

```
plot(f,20*log10(P{1}), 'k', f,20*log10(P{2}), 'r', f,20*log10(P{3}), 'b', 'LineWidth',1.5);
```

```
legend(names); title('Window Comparison'); xlim([50 170]); grid on;
```

```
%% Window Shapes
```

```
figure;
```

```
for i = 1:3
```

```
    subplot(3,1,i); plot(w{i}, 'LineWidth',1.5);
```

```
    title([names{i} ' Window']); grid on;
```

```
end
```

```
%% Peak Detection
```

```

for i = 1:3

    [pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);

    fprintf('\n%s Window Frequencies:\n',names{i});

    disp(loc{i});

end

%% Peak Marking (Hamming)

figure;

plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;

plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');

title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;

%% Main Peaks

fprintf('\nMain Spectral Peaks:\n');

for i = 1:3

    [~,idx] = max(P{i});

    fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));

end

```

Output:

Rectangular Window Frequencies:

100.0977 109.8633

Hamming Window Frequencies:

100.0977 109.8633

Hanning Window Frequencies:

100.0977 109.8633

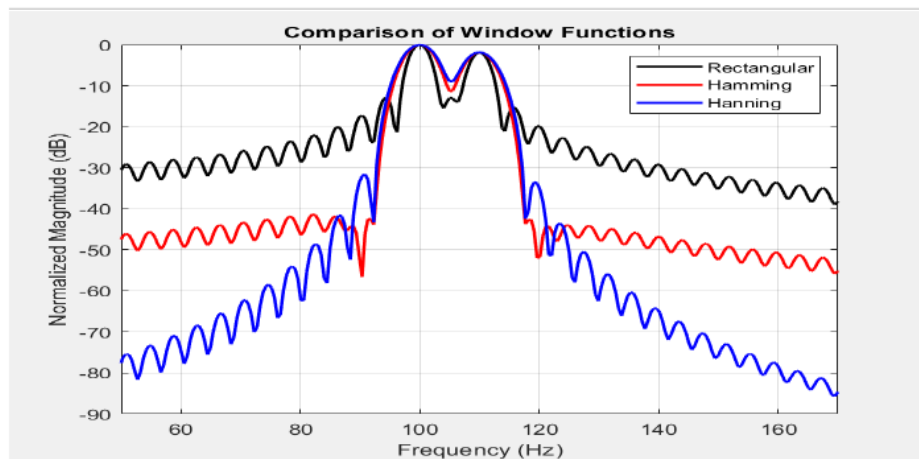
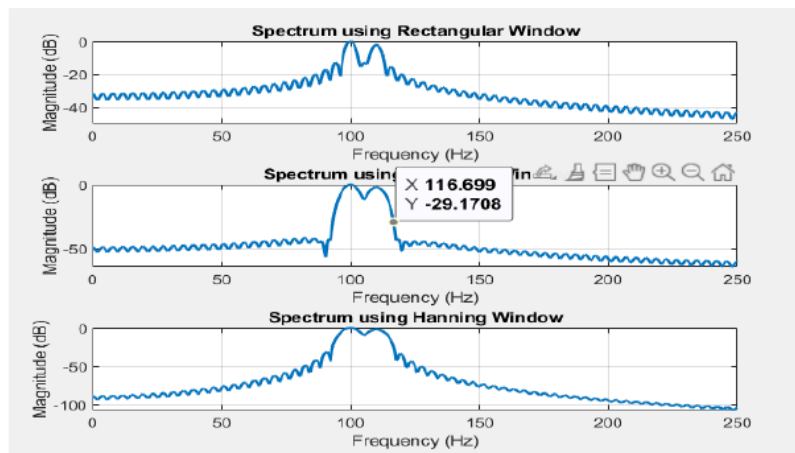
Main Spectral Peaks:

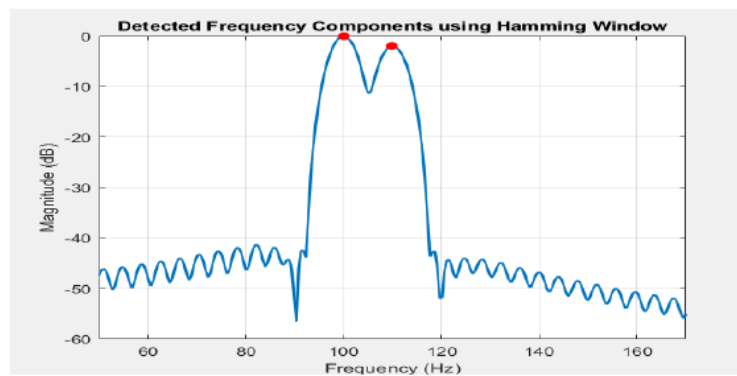
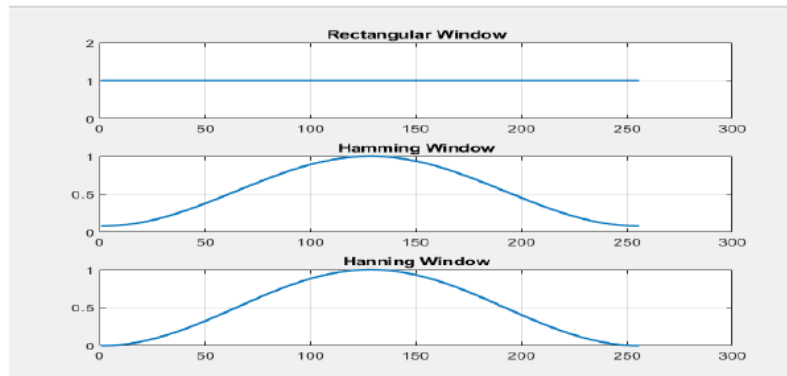
Rectangular Window Peak = 100.10 Hz

Hamming Window Peak = 100.10 Hz

Hanning Window Peak = 100.10 Hz

>>





Spectrum Estimation

Output

Detected Frequencies using Rectangular Window (Hz)
100.0977 109.8633

Detected Frequencies using Hamming Window (Hz)
100.0977 109.8633

Detected Frequencies using Hanning Window (Hz)
100.0977 109.8633

Result:

1. The spectrum of the given signal was successfully estimated using Rectangular, Hamming, and Hanning window functions in MATLAB.
2. The effect of window selection on spectral leakage was analyzed, showing that the Rectangular window exhibits higher leakage compared to Hamming and Hanning windows.
3. The main-lobe width and side-lobe levels were compared, where Hamming and Hanning windows provided better side-lobe suppression at the cost of slightly wider main-lobes.
4. The accuracy of frequency component detection was evaluated, and it was observed that windowing improves frequency resolution and reduces spectral distortion.

Practical Question 1:

Generate a discrete-time signal consisting of sinusoidal frequency components at 100 Hz and 250 Hz, sampled at 1000 Hz. Apply Rectangular, Hamming, and Hanning window functions and estimate the spectrum using FFT in MATLAB. Analyze the resulting spectral estimates and compare the effect of each window on spectral leakage, main-lobe width, side-lobe attenuation, and frequency component detection accuracy.

Task:

1. Generate a discrete-time signal containing sinusoidal components at 100 Hz and 250 Hz sampled at 1000 Hz.
2. Apply Rectangular, Hamming, and Hanning window functions to the generated signal.
3. Compute the FFT-based magnitude spectrum for each windowed signal and plot the results.
4. Compare and analyze the spectral characteristics of each window in terms of spectral leakage, main-lobe width, side-lobe attenuation, and frequency detection accuracy.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 1:

Generate a discrete-time signal consisting of two sinusoidal components at 120 Hz and 300 Hz, sampled at 2000 Hz. Apply Rectangular, Hamming, and Hanning window functions and estimate the spectrum using FFT in MATLAB. Compare and analyze the effect of each window on spectral leakage, main-lobe width, side-lobe attenuation, and the accuracy of frequency component identification.

Task:

1. Generate a discrete-time signal with frequency components of 120 Hz and 300 Hz sampled at 2000 Hz.
2. Apply Rectangular, Hamming, and Hanning window functions to the signal.
3. Compute the FFT-based spectrum for each windowed signal and plot the magnitude spectra.
4. Compare and analyze the spectral performance of each window in terms of leakage, main-lobe width, side-lobe levels, and frequency detection accuracy.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To compare spectral leakage, main-lobe width, side-lobe attenuation, and frequency detection using three windows.

Student Details

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